MindVault Protocol Specification: API Gateway Routing and Server-Sent Events
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Section 1: Gateway Routing Architecture and Endpoints
The gateway dispatches incoming POST /api/rag/stream requests to RAGService. RAGService triggers RetrievalSe
on PostgreSQL document_chunks using hybrid vector and keyword channels. SSE stream lifecycle emits standard
start -> retrieval_complete -> sources -> token* -> validation -> done.

Section 2: Connection Heartbeat and Abort Lifecycle
The gateway dispatches a comment heartbeat (: ping) every 15 seconds to prevent intermediate proxy timeout.
Client socket termination propagates an AbortSignal, terminating upstream LLM generators and marking the record

Section 3: Error Envelopes and Ingestion Format Boundaries
Supported extensions: .pdf, .docx, .txt, .md. Spreadsheets must be converted to Markdown tables.
Error event envelope emits code FILE_TOO_LARGE and transitions document to FAILED when payload exceeds 2

Section 4: Context Assembly Dual Budgeting Limits
Rate limit excess yields 429 Too Many Requests. ContextAssembler drops chunks exceeding maxContextTokens a
the 75% narrative floor and 25% graph ceiling policy. Chunks that do not fit within the token budget are omitted.